

08-Traceability-Matrix

Traceability Matrix — Leish! v2

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Traceability Matrix — Leish! v2

FIELD	VALUE
Document ID	LEISH-RTM-v2.0
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Purpose	Bidirectional trace: Business Requirement → SRS → FRS → API/UI → Data → Verification. Ensures no orphan requirement and no unvalidated feature.

1. End-to-End Trace (BR → FRS → Implementation → Test)

BR	BRS §	SRS- F	FRS IDS	APIS / PAGES	TABLES / LOGIC
BR-01 Public catalog discovery	6.1	F-02	FR-CAT-01	GET /api/artists , GET /api/catalog/artists , GET /artists , /studios , /	artists, studios + listArtists / AREAS_BY_STATE
BR-02 Filtered artists	6.1	F-02	FR-CAT-01, FR-CAT-03	GET /api/artists? state&area&bridal&nonBridal&budget&query	artistsQuerySchema , idx_artists_state_area
BR-03 Detail view	6.1	F-02	FR-CAT-02	GET /artists/[slug] , GET /studios/[slug] (RSC resolveArtist)	artists, studios, reviews via listEntityReviews
BR-04 Gated reviews	6.1	F-07	FR-REV-01, FR-REV-02	POST /api/artists/[slug]/reviews	reviews.booking_id UNIQUE + findReviewableBooking
BR-05 Atomic rating blend	6.1	F-07	FR-REV-03	(internal)	blendAggregate() single UPDATE ROUND(...,2)
BR-06 Registration	6.2	F-01	FR- AUTH-01, FR- AUTH-07	POST /api/auth/register , /register	users , registerSchema , crypt+pepper
BR-07 Email verification gate	6.2	F-01	FR- AUTH-05, FR- AUTH-02	GET /api/auth/verify-email , POST /api/auth/ resend-verification , /verify-email	email_verifications hashed token, users.email_verified , banner in /dashboard
BR-08 Password reset	6.2	F-01	FR- AUTH-06	POST /api/auth/forgot-password , POST /api/ auth/reset-password , /forgot-password , / reset-password	password_resets hashed 1h single-use
BR-09 Session 7d + revocation	6.2	F-01	FR- AUTH-03, FR- AUTH-04	POST /api/auth/login , POST /api/auth/logout , Cookie leish_session	sessions JTI, SESSION_TTL_SECONDS , verifySessionToken
BR-10 Export / delete (PDPA)	6.2	F-13	FR- COMP-01, FR- COMP-02	GET /api/me/export , DELETE /api/me? confirm=1 , /dashboard buttons	FK CASCADE / SET NULL
BR-11 Booking request with event details	6.3	F-03	FR- BOOK-01	POST /api/bookings , /artists/[slug]/book	bookingSchema , bookings + server-derived price , notifyBookingCreated
BR-12 Slot uniqueness	6.3	F-03	FR- BOOK-01	POST /api/bookings 409 path	uq_bookings_slot partial index + pre-check + race catc

BR	BRS §	SRS- F	FRS IDS	APIS / PAGES	TABLES / LOGIC
BR-13 Accept/ reject/ complete	6.3	F-03	FR- BOOK-03	<code>PATCH /api/bookings/[id] {accept\ reject\ complete\ cancel}</code>	<code>applyBookingTransition</code> state machine
BR-14 Quotation 24h + re- quote	6.3	F-04	FR-QUO-01, FR-QUO-02	<code>POST /api/bookings/[id]/quotation</code> , <code>/dashboard</code> quote block	<code>quotations</code> <code>status</code> <code>pending/superseded/expired</code> <code>paid</code> , <code>quotationTotal</code> , <code>QUOTATION_TTL_MS</code>
BR-15 Quotation display + balance due	6.3	F-03, F-04	FR- BOOK-04	<code>GET /api/bookings</code> <code>serializeBooking</code> (quotation,totalPrice,balanceDueDate,balanceAmount)	<code>getBookingFeeSen()</code> , <code>balanceDueDate = date -30</code>
BR-16 Pay deposit	6.4	F-05	FR-PAY-02	<code>POST /api/bookings/[id]/pay-fee</code>	<code>payments(type=deposit)</code> , <code>createBookingPayment</code> , Billplz <code>POST /bills</code>
BR-17 Pay balance T-3d	6.4	F-05	FR-PAY-03	<code>POST /api/bookings/[id]/pay-balance</code>	<code>payments(type=balance)</code> , <code>amount=total-fee</code>
BR-18 Webhook HMAC verified	6.4	F-05	FR-PAY-04	<code>POST /api/payments/webhook</code> + <code>X-Billplz-Signature</code>	<code>verifyBillplzSignature</code> timingSafe, <code>markBillPaid</code> , <code>handlePaymentPaid</code>
BR-19 Webhook routes + confirms / creates payout	6.4	F-05, F-06	FR-PAY-04, FR- PAYO-02	same webhook	<code>confirmOnFeePaid</code> vs <code>createPayoutForBooking</code> routed by <code>type</code>
BR-20 Commission computed	6.4	F-06	FR- PAYO-01	(internal, admin settings)	<code>platform_settings</code> , <code>computeCommission</code> , rate clamp 0..5000
BR-21 Payout settle/fail	6.4	F-06	FR- PAYO-03	<code>GET/PATCH /api/admin/payouts</code> + <code>/admin/payouts</code>	<code>payouts.status</code> <code>pending→settled/failed</code> , <code>settleable_at = event+24h</code>
BR-22 Refund balance only	6.4	F-05	FR-PAY-05	<code>POST /api/bookings/[id]/refund</code>	<code>refundBalancePayment</code> guard <code>type=balance</code> + <code>status=paid</code>
BR-23 Chat live	6.5	F-08	FR- MSG-01, FR-MSG-02	<code>GET/POST /api/bookings/[id]/messages</code> , <code>GET .../messages/stream</code> SSE	<code>messages</code> + <code>chat-bus</code> pub sub

BR	BRS §	SRS- F	FRS IDS	APIS / PAGES	TABLES / LOGIC
BR-24 Email preferences	6.5	F-11	FR-MAIL-03	<code>GET/PATCH /api/email/preferences</code>	<code>email_preferences</code> flags
BR-25 Email retries + outbox	6.5	F-11	FR-MAIL-01, FR-MAIL-02	<code>POST /api/bookings/*</code> email side-effects, <code>/dev/emails</code>	<code>email_outbox</code> , <code>email_retries</code> , <code>sendEmail</code> fallback
BR-26 Admin CRUD	6.6	F-10	FR-ADM-02	<code>GET/POST/PATCH/DELETE /api/admin/users</code> , <code>.../artists</code> , <code>.../studios</code> , <code>GET/PATCH .../bookings</code> , <code>GET .../payments</code> , <code>GET .../quotations</code> , <code>GET .../messages</code>	<code>artists, studios, bookings</code> , via <code>updateArtist</code>
BR-27 Audit trail	6.6	F-10	FR-ADM-03	all admin mutators	<code>admin_audit_log</code> + <code>logAdminAction()</code> + <code>idx_audit_log_*</code>
BR-28 Tunable settings	6.6	F-10, F-15	FR-ADM-04, FR-OPS-01	<code>GET/PATCH /api/admin/settings</code> , <code>/admin/settings</code>	<code>platform_settings</code> , <code>getBookingFeeSen()</code> etc + 30s cache
BR-29 Dashboard metrics	6.6	F-10	FR-ADM-02 (dashboard)	<code>GET /api/admin</code>	Aggregated counts <code>users by role</code> , <code>bookings by status</code> , <code>payments revenue</code>
BR-30 Structured logs + forwarding	6.7	F-12	FR-OPS-01 (NFR)	<code>src/server/logger.ts</code>	<code>pino</code> + <code>LOG_WEBHOOK_URL</code> batch 20/50ms
BR-31 Error reporting	6.7	F-12	(NFR-S)	<code>src/server/errors.ts</code>	<code>reportError</code> → Sentry / <code>ERROR_WEBHOOK_URL</code>
BR-32 Analytics events	6.7	F-12	FR-OPS-03	<code>src/server/agnost.ts</code> , <code>src/lib/agnost-client.ts</code> , <code>src/instrumentation.ts</code>	<code>agnost.begin/end</code> , <code>trackEvent</code>
BRL-*	BRS §7	SRS §6-7	—	—	Enforced by DB CHECK + code guards
NFR-P/R/S/U/M/O	BRS §10-12	SRS §7, NFR doc	—	Headers, CSP, rate-limit, <code>next.config.ts</code>	—

2. Business Rule → Test Matrix (BRL → Enforcement → Test)

BRL	RULE	ENFORCEMENT (DB/CODE)	TEST LOCATION / SPEC
BRL-01	Slot atomic	<code>uq_bookings_slot</code> partial index <code>src/server/db.ts:470</code> ; service guard <code>POST /api/bookings</code>	Booking concurrency test (cohort 2 par requests → 1 409)
BRL-02	Price authority	<code>serviceDef.price</code> server derive <code>src/app/api/bookings/route.ts:183</code> ; <code>quotationTotal()</code> server-only	Booking create test asserts custom price ignored
BRL-03	Sen integers	<code>INTEGER</code> columns + <code>Math.round</code> in <code>quotations.ts</code> / <code>settings.ts</code> / <code>payouts.ts</code>	Lint + payment math tests
BRL-04	Deposit non-refundable	<code>refundBalancePayment()</code> guard <code>type!</code> <code>=balance → throw</code> <code>src/server/payments.ts:322</code>	Refund test attempts deposit → error
BRL-05	Confirm only via webhook	<code>confirmOnFeePaid</code> called only in <code>handlePaymentPaid</code> <code>src/server/payments.ts:246</code> ; PATCH rejects <code>confirmed</code>	Webhook vs PATCH tests
BRL-06	24h quotation TTL	<code>QUOTATION_TTL_MS=86400000</code> <code>src/server/quotations.ts:13</code> ; <code>isQuotationExpired</code> lazy + sweep	Expiry test (mock now+25h → expired)
BRL-07	Review gated	<code>reviews.booking_id</code> UNIQUE + <code>findReviewableBooking</code> <code>src/server/catalog.ts:678</code>	Gate test: uncompleted → null; duplicate review → 409
BRL-08	Commission waiver	<code>computeCommission</code> waived branch <code>src/server/settings.ts:106</code>	Unit test total<waiver → 0
BRL-09	Commission cap	Clamp <code>min(5000,max(0,round(v)))</code> <code>src/server/settings.ts:77</code>	Cap test rate 6000 → 5000
BRL-10	Verification gate	Payment handler checks <code>email_verified</code> (strict flag)	Pay before verified → 403 when enabled
BRL-11	Profile claim scoping	<code>artist_profiles</code> UNIQUE + <code>GET /api/bookings</code> claimedId filter <code>src/app/api/bookings/route.ts:99</code>	Scoping test: artist sees only claimed
BRL-12	Settlement 24h	<code>settleable_at = eventDate+24h</code> <code>src/server/payouts.ts:81</code>	Payout date math test
BRL-13	Audit completeness	<code>logAdminAction()</code> in every admin mutator + <code>src/app/api/admin/list-routes.test.ts</code> coverage audit	Admin mutation test asserts audit row created
BRL-14	Rate limit	<code>src/server/rate-limit.ts</code> sliding window + 429 header	Rate-limit contract test
BRL-15	7d JTI revoke	<code>src/server/session.ts</code> <code>SESSION_TTL_SECONDS=604800</code> + <code>revokeSession</code> wins	Session revoke test

3. NFR → Gate Matrix

NFR ID	GATE TYPE	BLOCKING PR?	MEASUREMENT
NFR-P-01 (latency p95)	Bench on preview deploy; warn if > target	No (advisory for launch, blocking by v2.1)	k6 / Vercel p95
NFR-P-04 (pool, WAL)	Code review required fields present	Yes	Review checklist
NFR-R-02 (degradation)	Test: kill email/Billplz mock → still 200	Yes	Unit with mocked provider
NFR-S-03.03 (webhook sig)	Unit suite includes forged sig → 401	Yes (CI fails if missing)	Vitest
NFR-S-04 (headers)	<code>curl -I</code> assertions on <code>next.config.ts</code> headers	Yes	ZAP / curl
NFR-U-01 (a11y ≥90)	Lighthouse CI step <code>assert: a11y≥90</code>	Advisory pre-launch, blocking post-launch	Lighthouse CI
NFR-M-01 (lint/typecheck/coverage)	<code>quality-gate.yml</code> all green, coverage ≥80%	Yes	GHA required
NFR-O-02 (health)	<code>GET /api/health</code> 200 on PR deploy	Yes	Smoke fetch

4. Reverse Trace — Code → Requirement (Orphan Check)

IMPLEMENTATION	REQUIREMENT(S)	ORPHAN?
All <code>src/server/*</code> modules	See §1 rows	No — mapped
<code>src/components/*</code>	FRS §12 Dashboard, SRS §4.1 UI	Covered
<code>referrals</code> table + <code>referral_code</code> columns	BRS §4.2 deferred (referral MVP scaffold only) — no business rule yet; code is pre-requirement scaffolding for v2.2	Deferred — flag red if exposed before BR; tests exist but feature hidden
<code>products</code> type <code>src/lib/types.ts:89</code>	BRS §4.2 deferred e-commerce	Scaffold, not traced — ok if unused; must not gate launch
<code>src/lib/agnost-client.ts</code> client tracking	BR-32	Traced
<code>src/server/referral.ts</code> helpers	BRS deferred	Scaffold — see above; require BR before enabling rewards

Action: no undocumented production feature that bypasses a BR; scaffolds above are gated behind `referredBy` writes that only run when `referralCode` supplied (optional) — safe.

5. Change History Linkage

DOC VERSION	RTM CHANGE
v1 (2026-08-15)	BR-01..BR-16 only (single payment, no payouts)
v2 (2026-08-29)	Added BR-17..BR-25, BR-27..BR-32; added payments.type hybrid, payouts, messaging, admin audit, Agnost

6. Usage Notes

- **Before adding code:** add row to FRS → map to BR or create new BR (with product consent); never code from Jira alone.
- **Before shipping:** `git diff --stat` against this RTM — every changed table/endpoint must have a row or be scaffolding.
- **Before audit:** `npm run typecheck && npm run lint && npm test` + `src/app/api/admin/list-routes.test.ts` proves admin guard coverage.

This RTM is the single reference for reviewers (security, QA, compliance) to confirm every promised capability is implemented and tested.